HOTEL BOOKING CANCELLATION PREDICTION SYSTEM

Complete Machine Learning Project Report

Dataset: hotel_bookings.csv

Target: is_canceled (1 = canceled, 0 = not canceled)

Project phase: End-to-end workflow and Sprint 3 optimization

Prepared using the supplied project brief, dataset and trained model artifacts.

Executive Summary

This project builds a binary classification system to predict hotel booking cancellations. The supplied dataset contains **119,390 records** and **32 columns**; the cancellation rate is **37.0%**. The completed workflow includes data inspection, feature engineering, feature selection, hyperparameter tuning, final training, held-out evaluation and model serialization.

1. Problem Statement

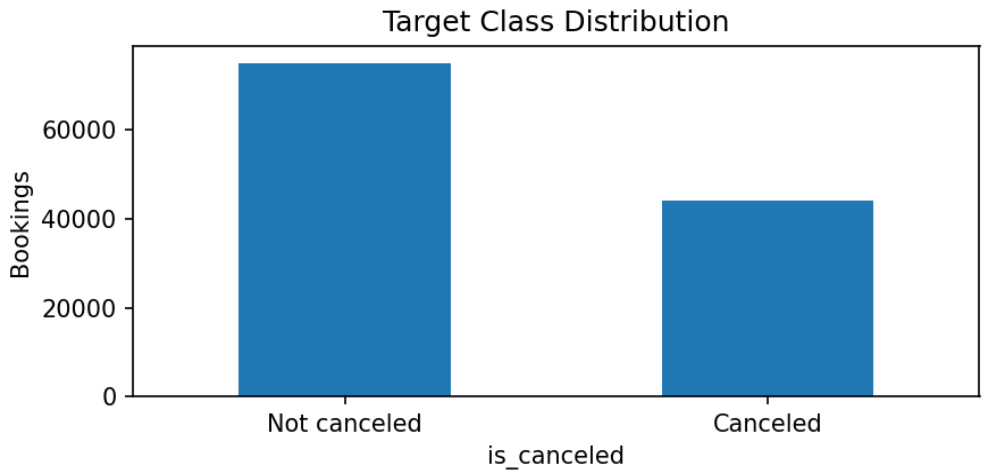
Hotel booking cancellations can lead to revenue loss, inefficient room allocation, overbooking or underutilization, and difficulty forecasting demand. The project asks whether a booking will be canceled based on booking details and customer-related information. Its intended business benefits include better room allocation, reduced revenue loss, improved pricing and overbooking strategies, and customer retention support.

2. Objective and Target Variable

Objective: predict whether a hotel booking is canceled. **Target column:** is_canceled. A value of 1 means canceled and 0 means not canceled. This is a supervised binary classification task.

3. Dataset Overview

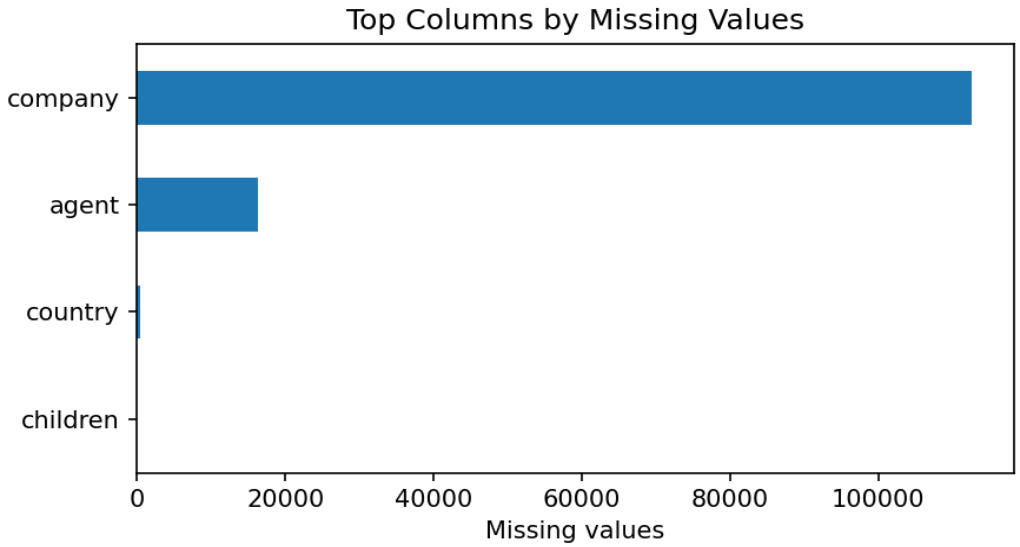
The CSV contains 119,390 rows and 32 columns. It includes numerical and categorical data. Fields described in the project brief include hotel, lead time, arrival date components, adults, children, babies, country, weekend/weeknight stays, meal, market segment, distribution channel, deposit type, ADR, previous cancellations and bookings, waiting-list days, room types, booking changes, agent/company and customer type.



Class counts — not canceled: 75,166; canceled: 44,224.

4. Data Quality and Preparation

Missing values were inspected. Numeric columns are imputed using the median and categorical columns using the most frequent value. Categorical features are one-hot encoded with unknown values handled safely and infrequent categories grouped. These transformations are placed inside the scikit-learn pipeline so they are learned from training data within cross-validation.



The chart shows the columns with the largest non-zero missing-value counts. Imputation is performed within the training pipeline.

Train/test split

A stratified 80/20 split was used: 95,512 rows for training and 23,878 for testing. Stratification preserves class proportions. The held-out test partition is reserved for final evaluation.

5. Feature Engineering

The following derived features were added:

Feature	Meaning
total_guests	Adults + children + babies.
total_nights	Weekend nights + week nights.
weekend_night_share	Weekend nights divided by total nights plus one.
previous_booking_count	Previous cancellations + previous non-canceled bookings.
previous_cancel_rate	Previous cancellations divided by previous booking count plus one.
special_requests_per_guest	Special requests divided by total guests plus one.
adr_per_guest	Average daily rate per guest, with guest count floored at one.
arrival_month_sin / cos	Cyclical encoding of arrival month / seasonality.

The arrival_date_month text field is replaced by sine/cosine month features. reservation_status and reservation_status_date are excluded because they can reveal the eventual outcome; assigned_room_type is also excluded to reduce reliance on potentially post-booking information.

6. Feature Selection

A correlation matrix is calculated using training rows only. One feature from numeric pairs with absolute correlation greater than 0.95 is removed. The recorded run removed: no features. Permutation importance is used to interpret feature contributions to ROC-AUC on a test sample; it is not used to iteratively tune the model on the test set.

7. Model Training and Hyperparameter Tuning

The final estimator is HistGradientBoostingClassifier. It is combined with a ColumnTransformer and preprocessing pipeline for numeric imputation, categorical imputation and one-hot encoding. RandomizedSearchCV uses ROC-AUC scoring and two-fold cross-validation on a stratified subset of the training data. The selected estimator is then refit on the full training partition.

Selected hyperparameters

Parameter	Value
max_leaf_nodes	15
max_iter	150
learning_rate	0.1
l2_regularization	0.0

Best cross-validation ROC-AUC: nan.

8. Final Evaluation on Unseen Test Data

The following metrics were computed on the held-out test partition at the default classification threshold.